



CS2300: File Structures and Introduction to Database Systems

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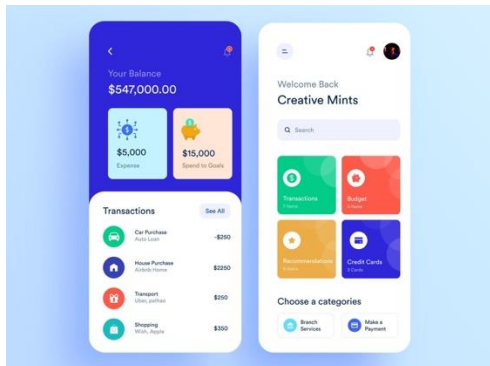
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Course Information & Policies

Project Head Start

- The database project has a UI component
 - UI = *user interface*
 - Command-line UI
 - Graphical UI (GUI) – opportunity to earn ***bonus credits***
- Examples of database applications and their GUIs



Mobile apps



Web apps



Desktop apps

What is a database?

(why are they an essential component of life in modern society?)

What is a database system?

(what might be the different types of database systems?)

Why this course?

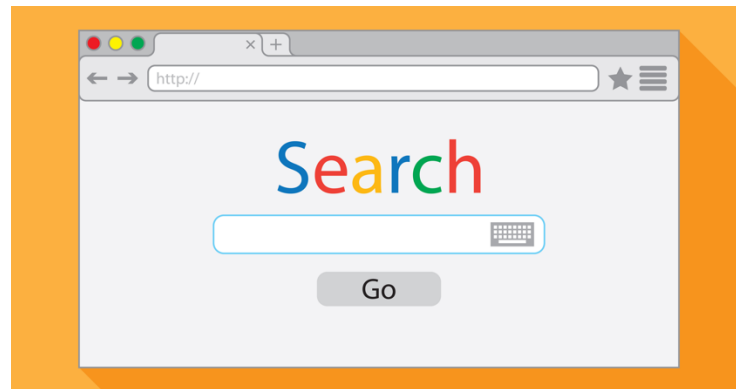
(do you want to work for Google, Amazon, and Yahoo?)

Question

- What's the difference between database systems and information retrieval systems?
 - Is one a subclass of another?



Is it a database-based app/service provider?



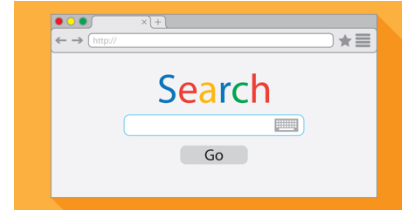
Is it an information retrieval system?

Databases and IR Systems: A Comparison



Database Systems

- **Structured data**
- Schema and data model driven
- **Structured query model and language**
- Rich metadata operations
- Query results are based on exact matching (always correct)



IR Systems

- **Unstructured data**
- No fixed schema
- **Free-from query models**
- Rich data operations
- Query results are based on approximate matching

Q1: Find the most expensive pair of shoes in the SHOES category.

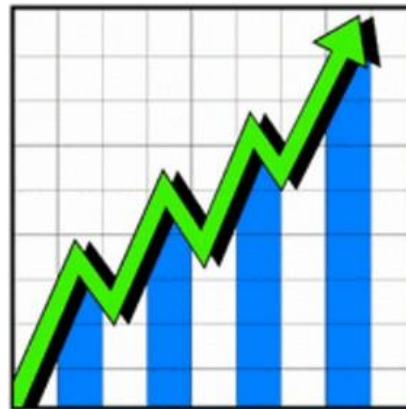
Q2: What is the current fashion trend for shoes in 2025?

Registration Forms



Databases touch all aspects of our lives

amazon[®]



 **Expedia**[®]

priceline.com[®]

Basic Definitions

- **Database:**
 - A collection of related data.
- **Data:**
 - Known facts that can be recorded and have an implicit meaning.
 - E.g. names of people
 - E.g. precise emotions of human?

However, a properly established database is often restricted with some additional implicit properties.

Implicit Properties for a Database

- A database represents some aspect of the real world, sometimes called the **miniworld**
 - Is a miniworld suppose to change?
- A database must be a logically coherent collection of data with inherent meaning
 - i.e. will a *random* assortment of data be counted as a database?
- A database is designed, built, and populated with data for a **specific purpose.**
 - Like an application...
 - It should have an intended *group of users*...

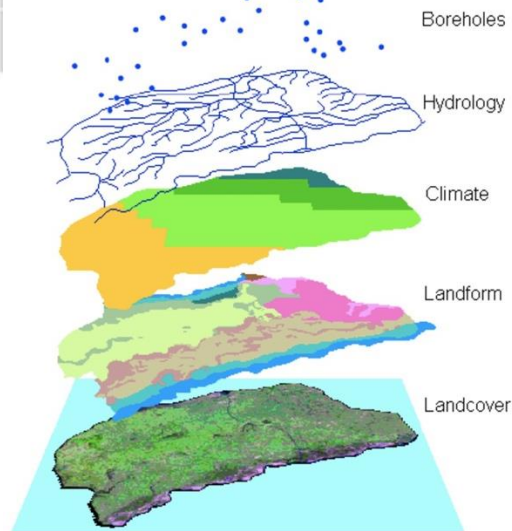
Fundamentals of Database Technology and Applications

Student ID	First Name	Last Name	Credits completed	GPA
34933	Victoria	Slater	36	
93759	Justin	McNeil	82	
20847	Jessica	Bennett	56	
65947	Michelle	Dolin	112	
24956	David	Price	57	
65692	Franklin	Mullins	13	
24271	Alissa	Lee		

Text/Numerical data



Multimedia data



GIS data

Social media data



An Example



A database that
stores student and
course information

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

data records

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

DBMSs for Databases

- **Database Management System (DBMS):**
 - A software package/ system to facilitate the creation and maintenance of a computerized database.
 - It **defines** a database:
 - such as specifying the data types and constraints of the data
 - It **constructs** a database:
 - Storing the data on some storage medium
 - It **manipulates** a database:
 - Query
 - Update
 - Even generate reports
 - **Sharing**
 - Allow multiple users and *programs* to access the database created simultaneously
- **Examples:** *MySQL, MongoDB, Google BigQuery, Neo4j*

Other Important DBMS Functions

- **Protection**

- *System protection*: against hardware or software malfunction or crashes
- *Security protection*: against unauthorized or malicious access

- **Data Presentation and Visualization**

- **Critical Maintenance**

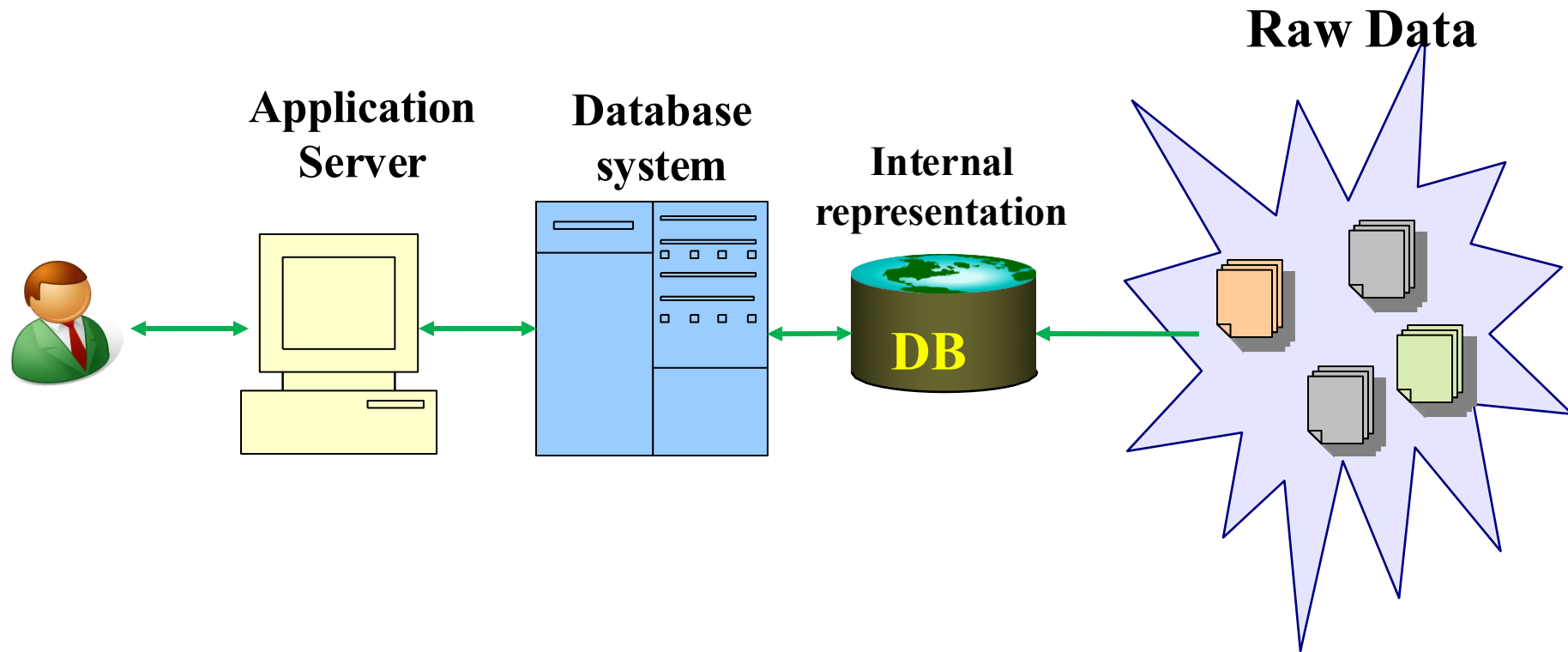
- Maintaining the database and associated programs over the lifetime of the database application
- Allow the system to evolve as requirements change over time

Database System Environment

In the previous example, there is an *application program* that interacts with the database

- **Application programs:** access the database by sending queries or requests for data to the DBMS
 - **Query:** mainly for retrieving data and information
 - **Transaction:** may cause some data to be read and some data to be written into the database

Simplified Database System Environment



YouTube - Broadcast Yourself - Windows Internet Explorer

http://www.youtube.com/

File Edit View Favorites Tools Help

Google codd wikipedia Search

Bookmarks Notebook Find Check AutoFill AutoLink

YouTube - Broadcast You... Gmail - Inbox (1725) - cohen... introduction database syste...

Sign Up QuickList (0) Help Sign In

Home Videos Users Community

Search Videos Search advanced Upload

Videos being watched right now...

Promoted Videos

Featured Videos

See More Featured Videos

Featured Most Viewed Most Discussed Top Favored

The Mysterious Explorations of Jasper Morello
From: monsterdistr...
Views: 57,134
★★★★★
Now playing in the YouTube Screening Room:
http://www.youtube.com/yt/screeningroom. Nomi (more)
26:08

Night Light
From: CurrencyFilms
Views: 45,852
★★★★★
NOTE: BEST WHEN VIEWED IN THE DARK.
This short film premiered at the '08 Tribeca Film (more)
07:15

There Are Monsters
From: blackdogfilms
Views: 361,211
★★★★★
Now playing in the YouTube Screening Room:
http://www.youtube.com/yt/screeningroom. On th (more)
09:56

Adam Sevani Halloween THRILLER Part 1
From: SevaniTube
Views: 342,652
★★★★★
A Tribute Video to Michael Jackson's 25th

Quantum Of Solace
IN THEATERS NOVEMBER 14
Theaters November 14th
From: QuantumOfSolace
Comments: 101
Login to rate video
★★★★★
954 ratings

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Captions and Subtitles
Add multi-track captions and subtitles to your videos

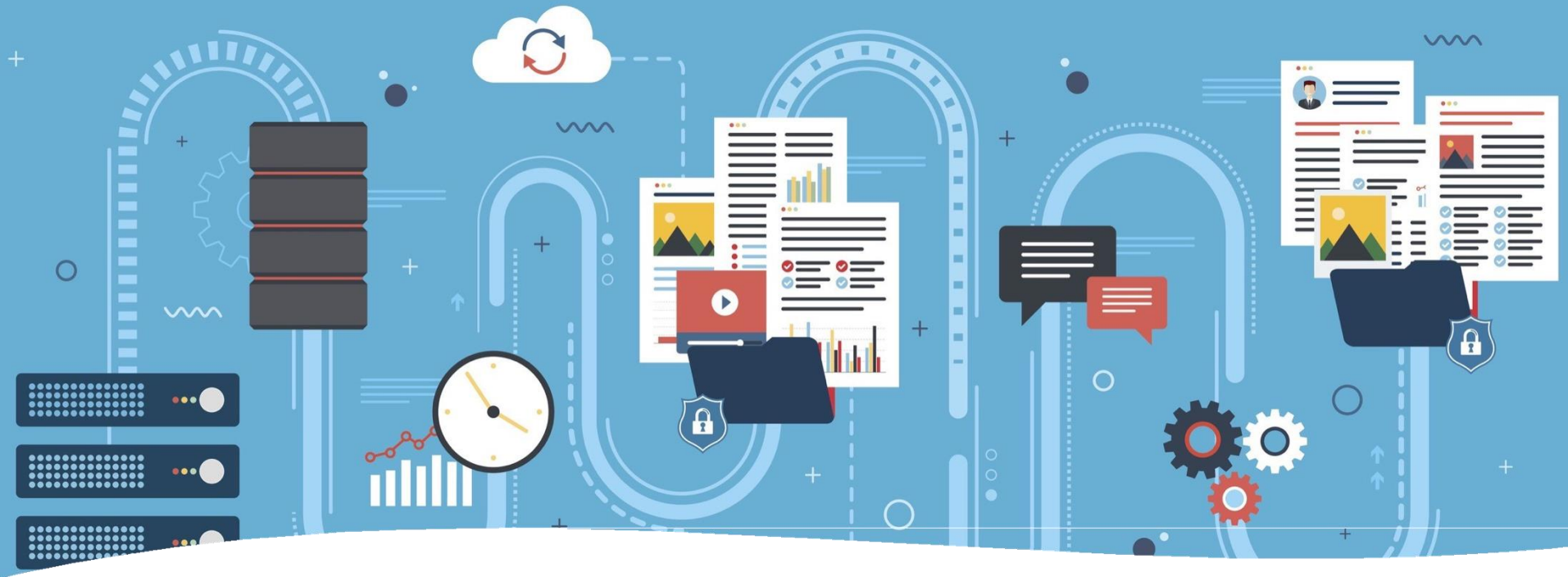
Video Annotations
Add interactive commentary and links to your videos

Auto Translate Now Available For Videos With Captions
We've added a new "auto translate" feature for

Start X inferno (XDMCP) X Find Message X Cygwin/X - inferno 2008 YouTube - Broadca... DBProvenance [Read... admin [Compatibility ... lecture1 [Compatibilit... EN 10:37 AM

Basic Definitions

- A **Database**:
 - A very large, integrated collection of related data.
- A **Database Management System (DBMS)** is:
 - A software system designed to store, manage, and facilitate access to databases.
- A **Database System**
 - The DBMS software together with the data itself. Sometimes, the applications are also included.



Characteristics of the Database Approach

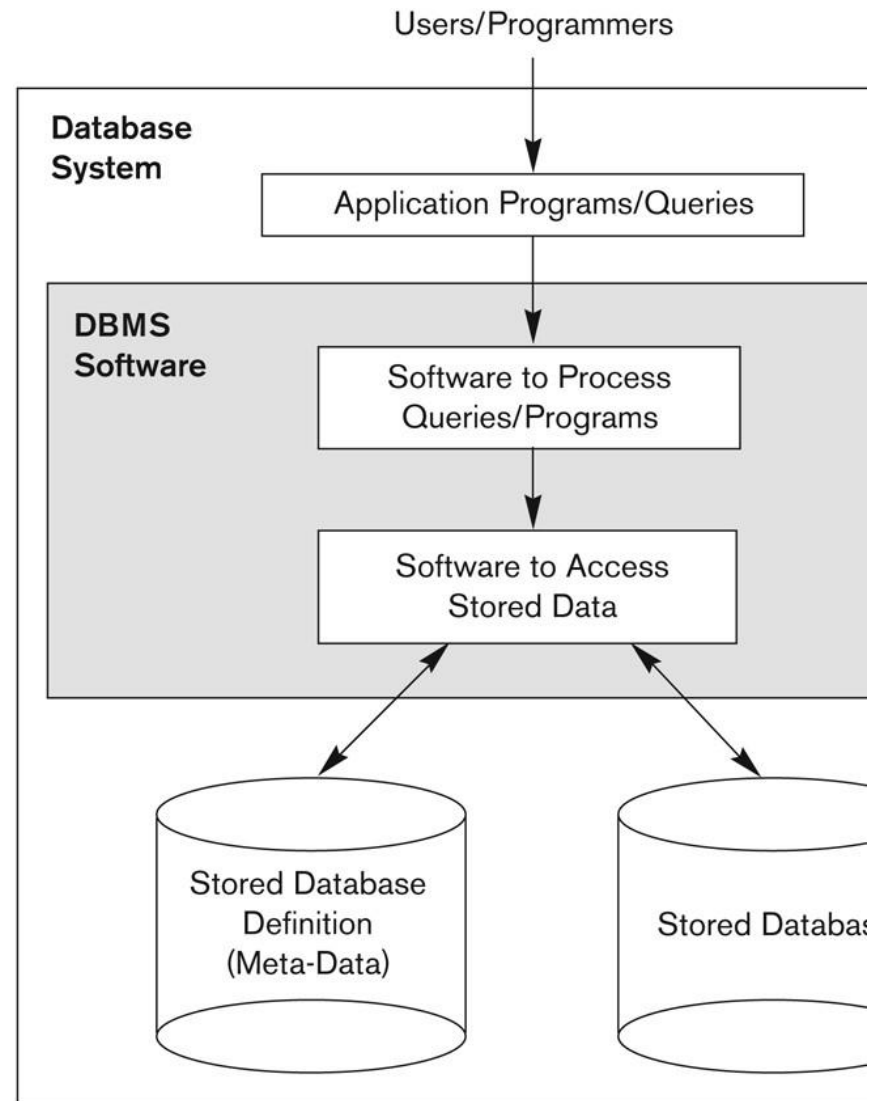
- The development and employment of using databases for information management can be seen as a new and improved way of doing things
- *The older way: file processing*
 - each user defines and implements the files needed for a specific software application
 - part of the job of programming the actual application

Characteristics of the Database Approach

Self-Describing Nature of a Database System

Meta-data:

- Information stored in the DBMS catalog that describes the structure of the primary database
- Structure of each file
- Data type and storage format of each data item
- Various constraints on the data



Self-Describing Nature of a Database System

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PREREQUISITE

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CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

A database that stores student and course information

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

An example of a db catalog for the student database

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

Characteristics of the Database Approach

Insulation between Programs and Data, and Data Abstraction

- **Program-Data Independence:** changes to the structures of data files stored in a DBMS will *not* impact any DBMS access programs
 - in traditional file processing, this would not be the case since the structure of data files is in majority embedded in the application programs

Characteristics of the Database Approach

Insulation between Programs and Data, and Data Abstraction

- **Program-Operation Independence:** the ability for users to easily define operations (i.e., **functions**) on the stored data as part of the database definitions.
- This allows application programs to operate on the data by calling these operations, **regardless** of how the operations are implemented.

Characteristics of the Database Approach

Insulation between Programs and Data, and Data Abstraction

- **Data Abstraction:** the characteristic that allows both program-data and program-operation independence in a database system.
- Data abstraction can be usually achieved through the means of some data models that are used to provide conceptual representation of data.
- The data model uses *logical* concepts, such as objects, their properties, and their interrelationships, to hide **storage** and **implementation** details that are not of interest to most database users.

Characteristics of the Database Approach

Support of Multiple Views of the Data

- **view:**
 - A subset of the database
 - or, *virtual data* that is **derived** from the database files (not explicitly stored)

Motivation:

- A database can have many types of ?_____? And a variety of distinct applications
- Each type may require a different ?_____? Or view of the database
- Some users may not need to be aware of whether the data they are seeing is ?_____? or derived

Support of Multiple Views of the Data

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CS3380	MATH2410
CS3320	CS1310

View 1

TRANSCRIPT

Student_name	Student_transcript				
	Course_number	Grade	Semester	Year	Section_id
Smith	CS1310	C	Fall	08	119
	MATH2410	B	Fall	08	112
Brown	MATH2410	A	Fall	07	85
	CS1310	A	Fall	07	92
	CS3320	B	Spring	08	102
	CS3380	A	Fall	08	135

(a)

View 2

COURSE_PREREQUISITES

Course_name	Course_number	Prerequisites
Database	CS3380	CS3320
		MATH2410
Data Structures	CS3320	CS1310

(b)

- Are the views useful?
- What determines if a view should be stored explicitly or derived?

Characteristics of the Database Approach

Sharing of Data and Multiuser Transaction Processing

- A *multiuser* DBMS must allow multiple users to access the database simultaneously
 - E.g. an **airline flight reservation** database system
- **concurrency-control**: DBMS managed software to ensure that multiuser updates (concurrent transactions) can be performed correctly
 - a fundamental requirement for OLTP (***online transaction processing***)
- **Key transaction properties**:
 - **isolation**: ensures that each transaction appears to execute in isolation from other transactions
 - **atomicity**: ensures that either all database-related operations in a transaction are executed or none are

Database Users – Actors on the Scene

Actors on the scene: the people whose jobs involve the day-to-day use of a large database system

- *What's the typical size of a database?*

Actors on the Scene	Role Responsibility
A.Database designers B.End users C.Database administrators D.System analysts, application programmers, and software engineers	_Oversee and manage the database system and resources _Choosing appropriate structures and representation for the data _Job requirements ask for daily database access for querying, updating, and generating reports _Must understand the full range of capabilities of the DBMS to satisfy the implementational requirements of end users

Database Career Opportunities

TABLE 1.3	DATABASE CAREER OPPORTUNITIES	
JOB TITLE	DESCRIPTION	SAMPLE SKILLS REQUIRED
Database Developer	Create and maintain database-based applications	Programming, database fundamentals, SQL
Database Designer	Design and maintain databases	Systems design, database design, SQL
Database Administrator	Manage and maintain DBMS and databases	Database fundamentals, SQL, vendor courses
Database Analyst	Develop databases for decision support reporting	QL, query optimization, data warehouses
Database Architect	Design and implementation of database environments (conceptual, logical, and physical)	DBMS fundamentals, data modeling, SQL, hardware knowledge, etc.
Database Consultant	Help companies leverage database technologies to improve business processes and achieve specific goals	Database fundamentals, data modeling, database design, SQL, DBMS, hardware, vendor-specific technologies, etc.
Database Security Officer	Implement security policies for data administration	DBMS fundamentals, database administration, SQL, data security technologies, etc.
Cloud Computing Data Architect	Design and implement the infrastructure for next-generation cloud database systems	Internet technologies, cloud storage technologies, data security, performance tuning, large databases, etc.
Data Scientist	Analyze large amounts of varied data to generate insights, relationships, and predictable behaviors	Data analysis, statistics, advanced mathematics, SQL, programming, data mining, machine learning, data visualization

Database Career Opportunities

Very large databases (VLDB), big data databases, in-memory databases, and cloud databases are the trending areas where database-related job opportunities are in hot demand. Which database career job interests you?



When to Use Customized Database Applications

- DBMSs provide a lot of advantages for database management

- *Can you name a few of them?*

Circumstances that a DBMS might not be worth it:

- Group discussion
 - What kind of projects/applications won't need a full-scale DBMS?
 - Volume of data: small vs large?
 - How much resources are needed to maintain a database system?
 - What if data sharing is not a requirement for you?

When Not to Use a DBMS

- Overhead of using a DBMS:
 - High initial investment in hardware, software, and training
 - Learning the generality that a DBMS provides for defining and processing data
 - Overhead for providing security, concurrency control, recovery, and integrity functions



When to Use Customized Database Applications

Circumstances that a DBMS might not be worth it:

- Simple database applications that are not expected to change at all
- Application programs that have stringent, real-time requirements
- Embedded systems with limited storage capacity, where a general-purpose DBMS would not be the best fit
- No use for multiple-user access to data

How a Database is Created

- Databases are a crucial component in the information system within any organization
- The design of a brand-new database starts off with the ***requirements specification and analysis*** phase
 - **Conceptual Design:**
 - Represent database requirements using some computerized tools
 - E.g. **Entity-Relationship Model**
 - **Logical Design:**
 - The translation of the conceptual design into a data model implemented in a commercial DBMS
 - **Physical Design:**
 - Executing implementational decisions for storing and accessing the database

Summary of Lecture 1 – Introduction to Databases

- What we learned?
- What you understood?
- What you don't quite understand?
- What do you want to have been covered but had not in this lecture?

